Musterwerk Automotive GmbH

Lastenheft: Electric Parking Brake (EPB)

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1.0	2026-10-29	B. Aksakal	Held-out test version with answer key

1 Introduction

This document specifies the Electric Parking Brake (EPB) of passenger cars (category M1). The EPB holds the parked vehicle with an electric actuator on the rear brakes and replaces the mechanical hand brake. Requirements are identified by IDs like EPB-R-010.

2 Apply and Hold

EPB-R-010 Apply time

When the driver pulls the EPB switch, the full clamping force shall be reached within 1.5 s.

EPB-R-011 Apply time limit

The full clamping force shall be reached no later than 1.0 s after the EPB switch is pulled.

EPB-R-012 Holding on slopes

The EPB shall hold the fully loaded vehicle on slopes of up to 30 %.

EPB-R-013 Maximum slope

The maximum slope on which the EPB shall hold the fully loaded vehicle is 20 %.

EPB-R-014 Applied indication

A red warning lamp shall be shown in the instrument cluster while the EPB is applied.

EPB-R-015 Release reaction

The EPB shall release fast when the driver presses the EPB switch.

EPB-R-016 Clamping on hot brakes

The EPB shall re-clamp with adequate force after the brake discs have cooled down.

3 Actuator and Power Supply

EPB-R-021 Actuator location

The EPB actuators shall be mounted on the rear brake calipers.

EPB-R-022 Dynamic braking

If the EPB switch is held while driving, the EPB shall decelerate the vehicle with at least 6 m/s².

EPB-R-024 Drive-away release condition

The automatic drive-away release shall only be available when the driver's seat belt is fastened.

EPB-R-026 Clamping force

Each EPB actuator shall generate a clamping force of at least 18 kN.

EPB-R-027 Automatic apply

The EPB shall apply automatically when the vehicle has been stationary with the engine off for 3 min.

EPB-R-028 Operating voltage

The EPB shall work at supply voltages from 9 V to 16 V.

EPB-R-029 Undervoltage fault

The EPB control unit shall store a fault when the supply voltage stays below 9 V for more than 500 ms.

4 Comfort and Display

EPB-R-030 Status message

The instrument cluster shall show the message "Parking brake applied" when the vehicle is switched off.

EPB-R-031 Actuator noise

The actuators shall operate as quietly as possible.

EPB-R-032 Release warning

An audible warning shall be clearly noticeable when the driver leaves the vehicle with the EPB released.

EPB-R-033 Fault indication

A yellow warning lamp shall be shown in the instrument cluster when an EPB fault is detected.

EPB-R-034 Service mode

The EPB shall provide a service mode that retracts the pistons for brake pad replacement.

5 Driver Assistance and Degradation

EPB-R-041 Dynamic braking limit

During dynamic braking with the EPB switch, the deceleration shall be limited to 4 m/s².

EPB-R-043 Drive-away release

The automatic drive-away release shall work regardless of the seat belt status.

EPB-R-045 Actuator force limit

The clamping force of each EPB actuator shall not exceed 1500 daN.

EPB-R-047 Automatic apply timing

The EPB shall apply automatically after the vehicle has stood still with the engine off for 120 s.

EPB-R-048 Robustness

The EPB shall work safely in all foreseeable situations.

EPB-R-049 Fault recovery

After a temporary undervoltage, the EPB shall be available again within a short time.

EPB-R-050 Manual release

The EPB shall only be released manually when the brake pedal is pressed.